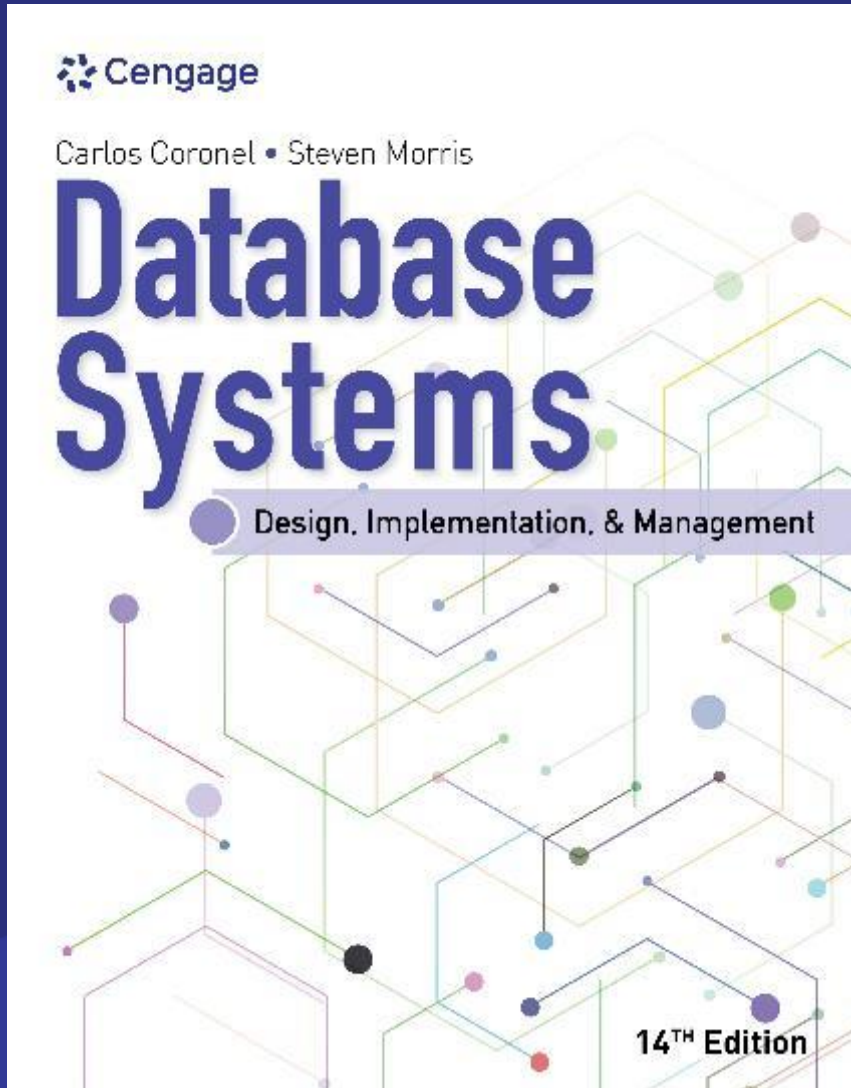




Database Systems: Design, Implementation, and Management, 14e

Module 4: Entity Relationship (ER) Modeling

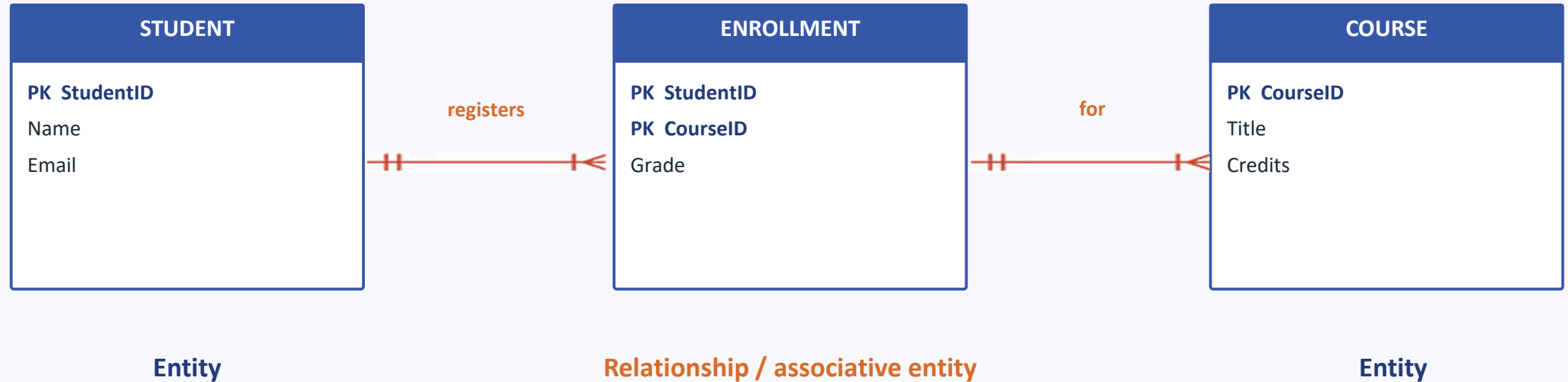
Chapter Objectives

By the end of this chapter, you should be able to:

1. Identify the main characteristics of entity relationship components
2. Describe how relationships between entities are defined
3. Explain how ERD components affect database design and implementation
4. Describe how real-world database design often requires the reconciliation of conflicting goals

ER Model —Recap

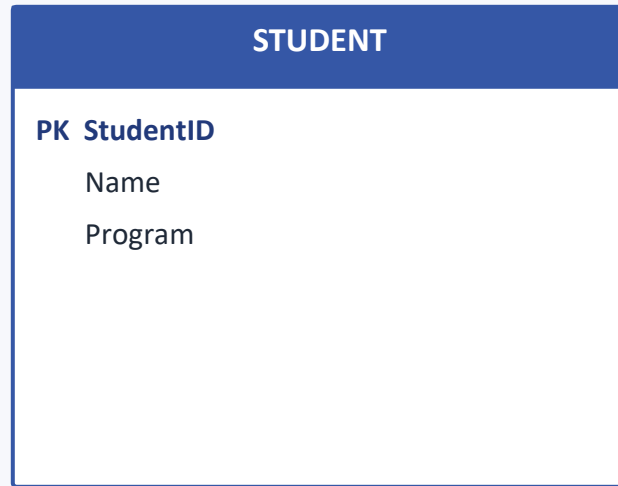
RECAP



Remember

Entity → table conceptually **entity instance** → one row **attribute** → column.

Entity vs. Entity Instance



Example rows (entity instances)

StudentID	Name	Program
101	Sara Ahmed	Computer Science
102	Justin Brisson	Networking
103	Maya Chen	Computer Science

Each row is one STUDENT occurrence.

ENTITY

The type/set of things we store data about.

Naming tip

Entity names are usually nouns: STUDENT, ORDER, PRODUCT, EMPLOYEE.

Attributes: What Describes an Entity?

ATTRIBUTES

Required

Must have a value.

Example

StudentID

Optional

May be left empty.

Example

MiddleName, SecondaryPhone

Identifier

Uniquely identifies each instance.

Example

StudentID

Domain

Allowed values for an attribute.

Example

Credits $\in \{1,2,3,4\}$

Design question

For every attribute, ask: Is it required? What values are valid? Can it uniquely identify a row?

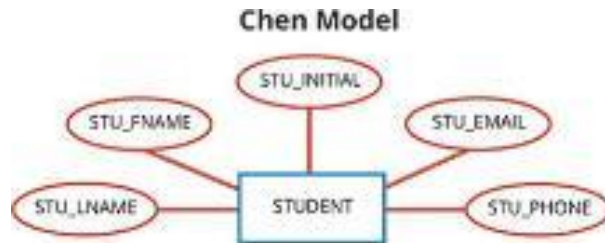
Attribute Types — with Concrete Examples

Type	Meaning	Example
Simple	Cannot be meaningfully split	FirstName
Composite	Can be divided into smaller parts	Address → Street, City, PostalCode
Single-valued	One value per entity instance	StudentID
Multivalued	Several values may exist	PhoneNumber(s)
Derived	Calculated from other stored data	Age from DateOfBirth

Implementation tip

Avoid storing a true multivalued attribute in one field.

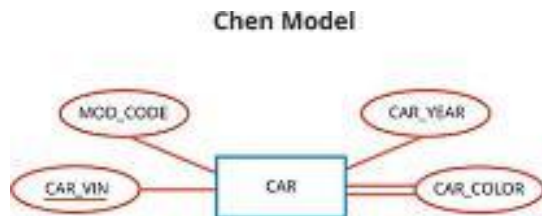
Attribute Types



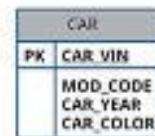
Crow's Foot Model



- The Attributes of the STUDENT Entity: Chen and Crow's Foot

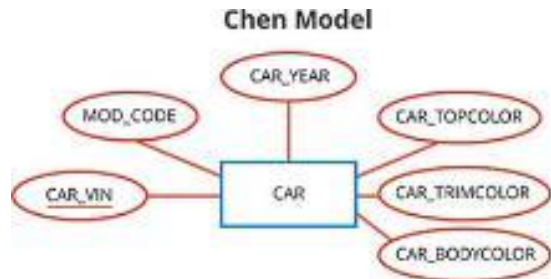


Crow's Foot Model

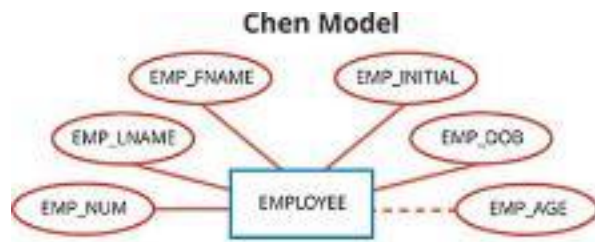


- A Multivalued Attribute in An Entity

Attribute Types



➤ Splitting the Multivalued Attribute into New Attributes



➤ Depiction of a Derived Attribute

Relationships

- The entities that participate in a relationship are also known as participants and each relationship is identified by a name that describes the relationship
- The relationship name is an active or passive verb
- Relationships between entities always operate in both directions

Connectivity vs. Cardinality

CARDINALITY

They are related, but they answer different questions.

Connectivity

What general relationship type is it?

1:1 • 1:M • M:N

Example: one professor can teach many classes → 1:M

Cardinality

What are the minimum and maximum counts?

(1,1) • (0,N) • (1,N)

Example: a professor may teach 0 to 4 classes → (0,4)

Connectivity = type **Cardinality** = exact participation limits.

Minimum / Maximum Cardinality Cheat Sheet

CROW'S FOOT

Read the symbol nearest each entity.

0..1

Optional one

A student may have zero or one locker.

1..1

Exactly one

Each class belongs to exactly one course.

0..*

Optional many

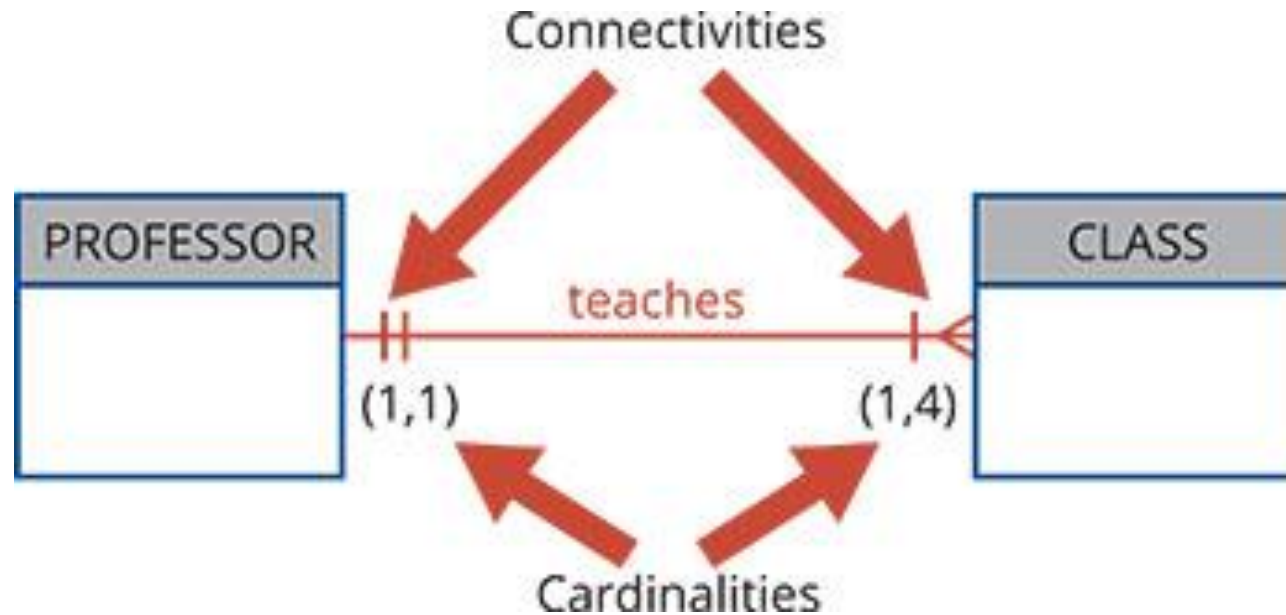
A professor may teach zero or many classes.

1..*

At least one

An order must contain one or more order lines.

Connectivity and Cardinality



Which side is optional? Which side is mandatory? Where would the foreign key normally be implemented?

Existence Dependence

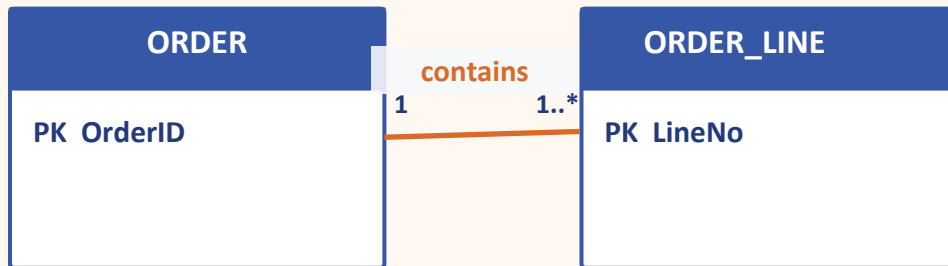
- An entity is said to be **existence-dependent** if it can exist in the database only when it is associated with another related entity occurrence
 - An entity is existence-dependent if it has a mandatory foreign key
- If an entity can exist apart from all of its related entities, it is **existence-independent**
 - It is referred to as a **strong entity** or **regular entity**

Existence Dependence

DEPENDENCE

Existence-dependent

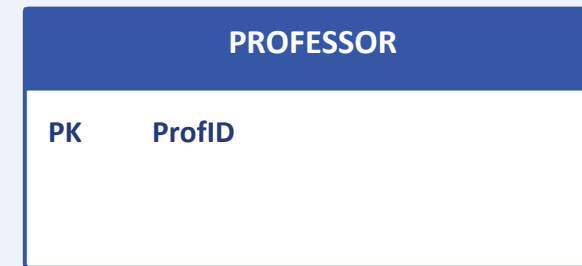
A child row cannot exist unless its parent exists.



No **ORDER_LINE** without an **ORDER**.

Existence-independent

The entity can exist even when no relationship currently exists.



A professor can exist before being assigned a class.

Relationship Strength (1 of 3)

- A relationship can be classified by whether the **parent's primary key becomes part of the child's primary key.**
- **Weak Relationship:** The parent's primary key appears in the child as a **foreign key only.**
- **Strong Relationship:** The parent's primary key appears in the child as a **foreign key AND as part of the child's primary key.**

Relationship Strength (2 of 3)

Weak Relationship

- The parent key appears in the child as a foreign key only.
- It is not part of the child's primary key.
- The child has its own independent primary key.

FK only → Weak

Strong Relationship

- The parent key appears in the child as a foreign key.
- It is also part of the child's primary key.
- The parent key helps identify each child record.

FK + part of PK → Strong

Note: relationship strength is different from whether an entity itself is weak or strong.

Example: Weak Relationship

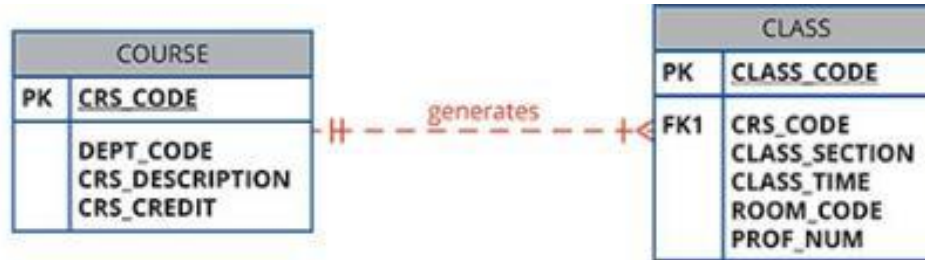


Table name: COURSE

Database name: Ch04_TinyCollege

CRS_CODE	DEPT_CODE	CRS_DESCRIPTION	CRS_CREDIT
ACCT-211	ACCT	Accounting I	3
ACCT-212	ACCT	Accounting II	3
CIS-220	CIS	Intro. to Microcomputing	3
CIS-420	CIS	Database Design and Implementation	4
MATH-243	MATH	Mathematics for Managers	3
QM-261	CIS	Intro. to Statistics	3
QM-362	CIS	Statistical Applications	4

Table name: CLASS

CLASS_CODE	CRS_CODE	CLASS_SECTION	CLASS_TIME	ROOM_CODE	PROF_NUM
10012	ACCT-211	1	MAF 8:00-8:50 a.m.	BUS311	105
10013	ACCT-211	2	MAF 9:00-9:50 a.m.	BUS200	105
10014	ACCT-211	3	TTh 2:30-3:45 p.m.	BUS252	342
10015	ACCT-212	1	MAF 10:00-10:50 a.m.	BUS311	301
10016	ACCT-212	2	Th 6:00-8:40 p.m.	BUS252	301
10017	CIS-220	1	MAF 9:00-9:50 a.m.	KLR209	228
10018	CIS-220	2	MAF 9:00-9:50 a.m.	KLR211	114
10019	CIS-220	3	MAF 10:00-10:50 a.m.	KLR209	228
10020	CIS-420	1	vW 6:00-8:40 p.m.	KLR209	162
10021	QM-261	1	MAF 8:00-8:50 a.m.	KLR200	114
10022	QM-261	2	TTh 1:00-2:15 p.m.	KLR200	114
10023	QM-362	1	MAF 11:00-11:50 a.m.	KLR200	162
10024	QM-362	2	TTh 2:30-3:45 p.m.	KLR200	162
10025	MATH-243	1	Th 6:00-8:40 p.m.	DRE155	325

Why is weak?

- CLASS has its own primary key: ClassID.
- CourseID is stored in CLASS as an FK.
- CourseID is not part of the CLASS primary key.

CLASS PK = ClassID
CourseID = FK only

Example: Strong Relationship

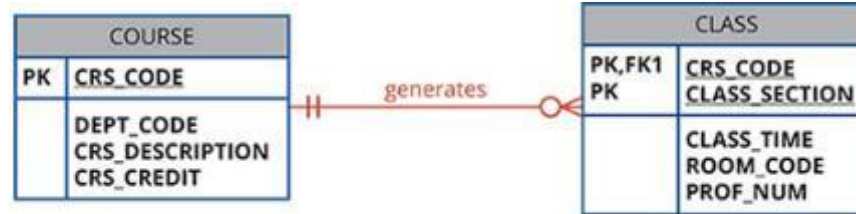


Table name: COURSE

CRS_CODE	DEPT_CODE	CRS_DESCRIPTION	CRS_CREDIT
ACCT-211	ACCT	Accounting I	3
ACCT-212	ACCT	Accounting II	3
CIS-220	CIS	Intro. to Microcomputing	3
CIS-420	CIS	Database Design and Implementation	4
MATH-243	MATH	Mathematics for Managers	3
GM-261	CIS	Intro. to Statistics	3
GM-362	CIS	Statistical Applications	4

Database name: Ch04_TinyCollege_Alt

Table name: CLASS

CRS_CODE	CLASS_SECTION	CLASS_TIME	ROOM_CODE	PROF_NUM
ACCT-211	1	MWF 8:00-8:50 a.m.	BUS311	105
ACCT-211	2	MWF 9:00-9:50 a.m.	BUS200	105
ACCT-211	3	TTh 2:30-3:45 p.m.	BUS252	342
ACCT-212	1	MWF 10:00-10:50 a.m.	BUS311	301
ACCT-212	2	Th 6:00-8:40 p.m.	BUS252	301
CIS-220	1	MWF 9:00-9:50 a.m.	KLR209	228
CIS-220	2	MWF 9:00-9:50 a.m.	KLR211	114
CIS-220	3	MWF 10:00-10:50 a.m.	KLR209	228
CIS-420	1	vV 6:00-8:40 p.m.	KLR209	162
MATH-243	1	Th 6:00-8:40 p.m.	DRE155	325
GM-261	1	MWF 8:00-8:50 a.m.	KLR200	114
GM-261	2	TTh 1:00-2:15 p.m.	KLR200	114
GM-362	1	MWF 11:00-11:50 a.m.	KLR200	162
GM-362	2	TTh 2:30-3:45 p.m.	KLR200	162

Why is it identifying?

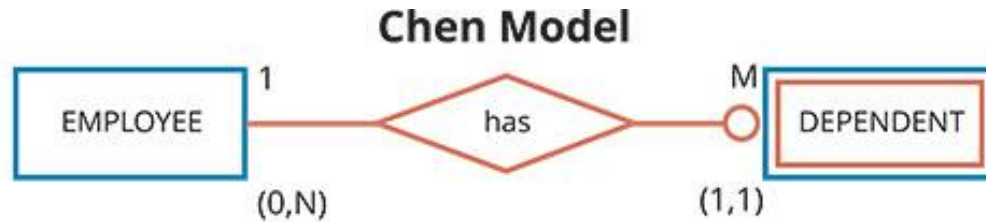
- CLASS is identified by CourseID + SectionNo.
- CourseID comes from the parent COURSE.
- CourseID is both a FK and part of the CLASS PK.

CLASS PK = (CourseID, SectionNo)

Weak Entities (1 of 3)

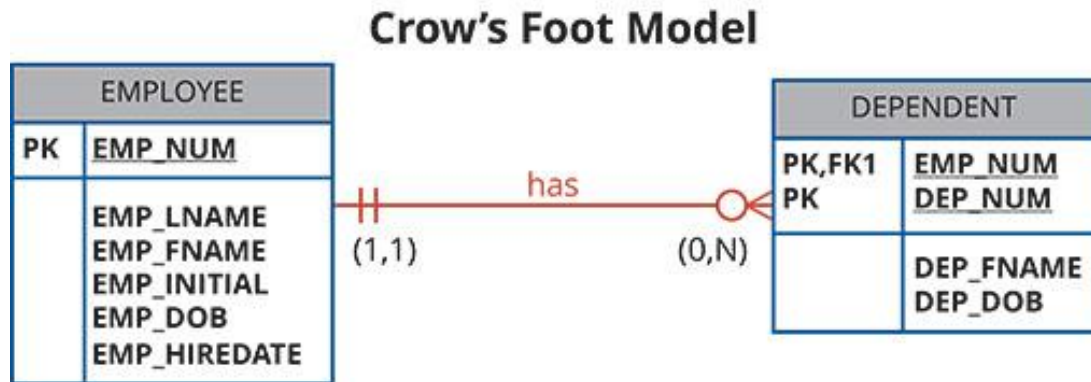
- A **weak entity** is one that meets the following two conditions:
 - The entity is existence-dependent
 - The entity has a primary key that is partially or totally derived from parent entity in the relationship
- The database designer usually determines whether an entity can be described as weak based on business rules

Weak Entities (2 of 3)



EMP_NUM
 EMP_LNAME
 EMP_FNAME
 EMP_INITIAL
 EMP_DOB
 EMP_HIREDATE

EMP_NUM
DEP_NUM
 DEP_FNAME
 DEP_DOB



A Weak Entity in an ERD

Weak Entities (3 of 3)

Table name: EMPLOYEE

Database name: Ch04_ShortCo

EMP_NUM	EMP_LNAME	EMP_FNAME	EMP_INITIAL	EMP_DOB	EMP_HIREDATE
1001	Callifante	Jeanine	J	12-Mar-84	25-May-97
1002	Smithson	William	K	23-Nov-70	28-May-97
1003	Washington	Herman	H	15-Aug-68	28-May-97
1004	Chen	Lydia	B	23-Mar-74	15-Oct-98
1005	Johnson	Melanie		28-Sep-66	20-Dec-98
1006	Ortega	Jorge	G	12-Jul-79	05-Jan-02
1007	O'Donnell	Peter	D	10-Jun-71	23-Jun-02
1008	Brzenski	Barbara	A	12-Feb-70	01-Nov-03

Table name: DEPENDENT

EMP_NUM	DEP_NUM	DEP_FNAME	DEP_DOB
1001	1	Annelise	05-Dec-97
1001	2	Jorge	30-Sep-02
1003	1	Suzanne	25-Jan-04
1006	1	Carlos	25-May-01
1008	1	Michael	19-Feb-95
1008	2	George	27-Jun-98
1008	3	Katherine	18-Aug-03

Figure 4.12 A Weak Entity in a Strong Relationship

Relationship Participation

- **Optional Participation:** An entity **can exist without** being related to another entity.
- **Mandatory Participation:** An entity **must be related** to another entity.
- **Easy way to remember:**
 - Optional = **may have a relationship**
 - Mandatory = **must have a relationship**

Relationship Participation: Optional vs. Mandatory

Look at the minimum cardinality.

Optional participation

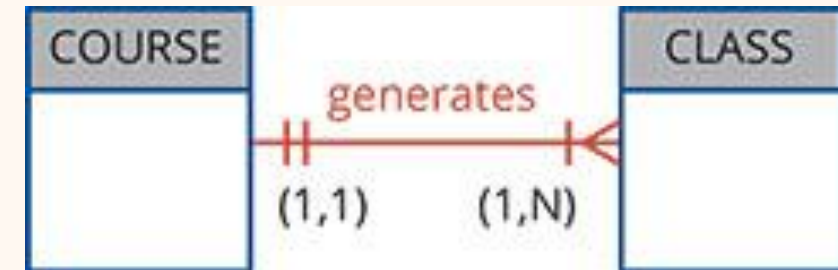
Minimum = 0



- An entity **may participate** in the relationship.
- A professor may teach **0 to 3 classes**.

Mandatory participation

Minimum = 1

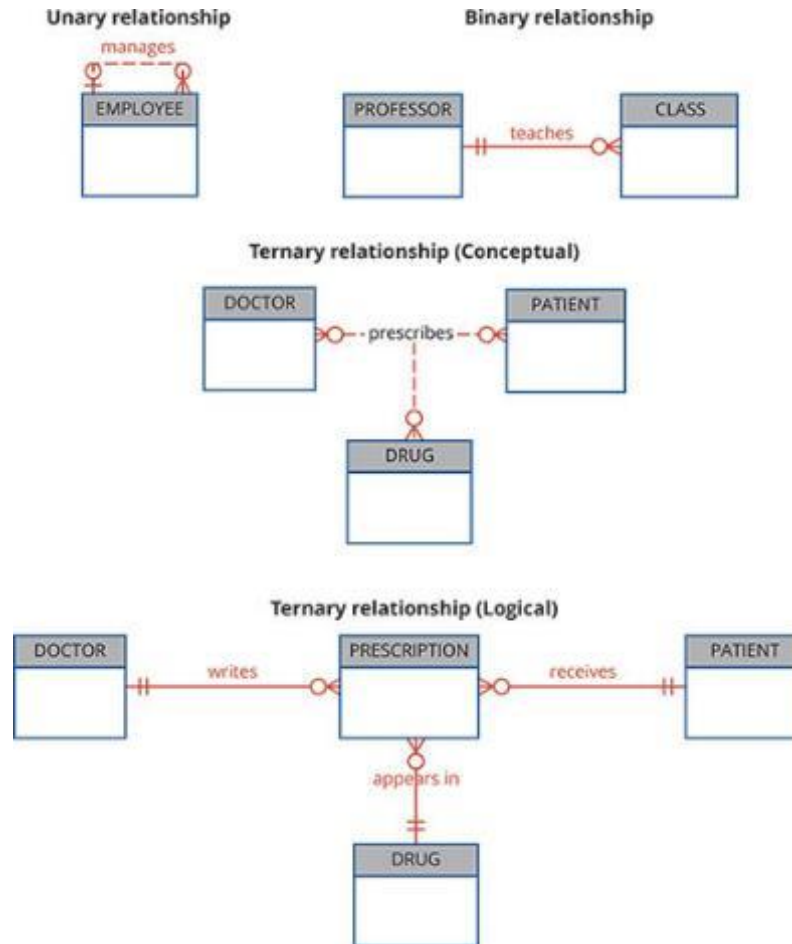


- An entity **must participate** in the relationship.
- A course must have **at least one class**.

Relationship Degree (1 of 3)

- A **relationship degree** indicates the number of entities or participants associated with a relationship
- A **unary relationship** exists when an association is maintained within a single entity
- A **binary relationship** exists when two entities are associated
- A **ternary relationship** exists when three entities are associated
- A **recursive relationship** is a relationship within a single entity type

Relationship Degree (2 of 3)



Three Types of Relationship Degree

Relationship Degree (3 of 3)

Database name: Ch04_Clinic

Table name: DRUG

DRUG_CODE	DRUG_NAME	DRUG_PRICE
AF15	Atgapon-15	25.00
AF25	Atgapon-25	35.00
DRO	Droalene Chloride	111.89
DRZ	Druzocholol Cryptolene	10.99
KO15	Kolabar Oxylhexalene	65.75
OLE	Oleander-Drizapan	123.95
TRYP	Tryptolac Heptadimetric	79.45

Table name: PATIENT

PAT_NUM	PAT_TITLE	PAT_LNAME	PAT_FNAME	PAT_INITIAL	PAT_DOB	PAT_AREACODE	PAT_PHONE
100	Mr.	Kolmycz	George	D	15-Jun-1942	615	324-5456
101	Mrs.	Lewis	Rhonda	G	19-Mar-2005	615	324-4472
102	Mr.	Vandam	Rhett		14-Nov-1958	901	675-8993
103	Mrs.	Jones	Anne	M	16-Oct-1974	615	890-3456
104	Mr.	Lange	John	P	08-Nov-1971	901	504-4430
105	Mr.	Williams	Robert	D	14-Mar-1975	615	890-3220
106	Mrs.	Smith	Jeanine	K	12-Feb-2003	615	324-7883
107	Mr.	Dante	Jorge	D	21-Aug-1974	615	890-4567
108	Mr.	Wiesenbach	Paul	R	14-Feb-1966	615	897-4358
109	Mr.	Smith	George	K	18-Jun-1961	901	504-3339
110	Mrs.	Genkazi	Leighla	W	19-May-1970	901	569-0093
111	Mr.	Washington	Rupert	E	03-Jan-1966	615	890-4925
112	Mr.	Johnson	Edward	E	14-May-1961	615	890-4387
113	Mrs.	Smythe	Melanie	P	15-Sep-1970	615	324-9006
114	Mrs.	Brandon	Marie	G	02-Nov-1932	901	882-0845
115	Mrs.	Saranda	Hermine	R	25-Jul-1972	615	324-5505
116	Mr.	Smith	George	A	08-Nov-1965	615	890-2904

Table name: DOCTOR

DOC_ID	DOC_LNAME	DOC_FNAME	DOC_INITIAL	DOC_SPECIALTY
29827	Sanchez	Julio	J	Dermatology
32445	Jorgensen	Annelise	G	Neurology
33456	Korenzki	Anatoly	A	Urology
33989	LeGrande	George		Pediatrics
34409	Washington	Dennis	F	Orthopaedics
36221	McPherson	Katye	H	Dermatology
36712	Dreifag	Herman	G	Psychiatry
38995	Minh	Tran		Neurology
40004	Chin	Ming	D	Orthopaedics
40028	Feinstein	Denise	L	Gynecology

Table name: PRESCRIPTION

DOC_ID	PAT_NUM	DRUG_CODE	PRES DOSAGE	PRES DATE
32445	102	DRZ	2 tablets every four hours -- 50 tablets total	12-Nov-22
32445	113	OLE	1 teaspoon with each meal -- 250 ml total	14-Nov-22
34409	101	KO15	1 tablet every six hours -- 30 tablets total	14-Nov-22
36221	109	DRO	2 tablets with every meal -- 60 tablets total	14-Nov-22
38995	107	KO15	1 tablet every six hours -- 30 tablets total	14-Nov-22

The Implementation of a Ternary Relationship

Knowledge Check Activity

What two conditions must be met before an entity can be classified as a weak entity?

Knowledge Check Activity: Answer

What two conditions must be met before an entity can be classified as a weak entity?

- **Answer:** To be classified as a weak entity, ***two conditions must be met:***
 - The entity must be existence-dependent on its parent entity.
 - The entity must inherit at least part of its primary key from its parent entity.

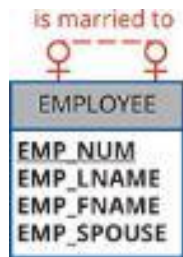
Recursive Relationships (1 of 2)

- A recursive relationship can exist between occurrences of the same entity set
 - Naturally, such a condition is found within a unary relationship
- One common pitfall when working with unary relationships is to confuse participation with referential integrity
 - Similar because they are both implemented through constraints on the same set of attributes

Recursive Relationships (2 of 2)



➤ An ER Representation of Recursive Relationships



Database name: Ch04_PartCo

Table name: EMPLOYEE_V1

EMP_NUM	EMP_LNAME	EMP_FNAME	EMP_SPOUSE
345	Ramirez	James	347
346	Jones	Anne	349
347	Ramirez	Louise	345
348	Delaney	Robert	
349	Shapiro	Anton	346

➤ The 1:1 Recursive Relationship “EMPLOYEE Is Married to EMPLOYEE”



Database name: Ch04_PartCo

Table name: EMPLOYEE_V2

EMP_CODE	EMP_LNAME	EMP_MANAGER
101	Waddell	102
102	Orincon	
103	Jones	102
104	Rebello	102
105	Robertson	102
106	Deltona	102

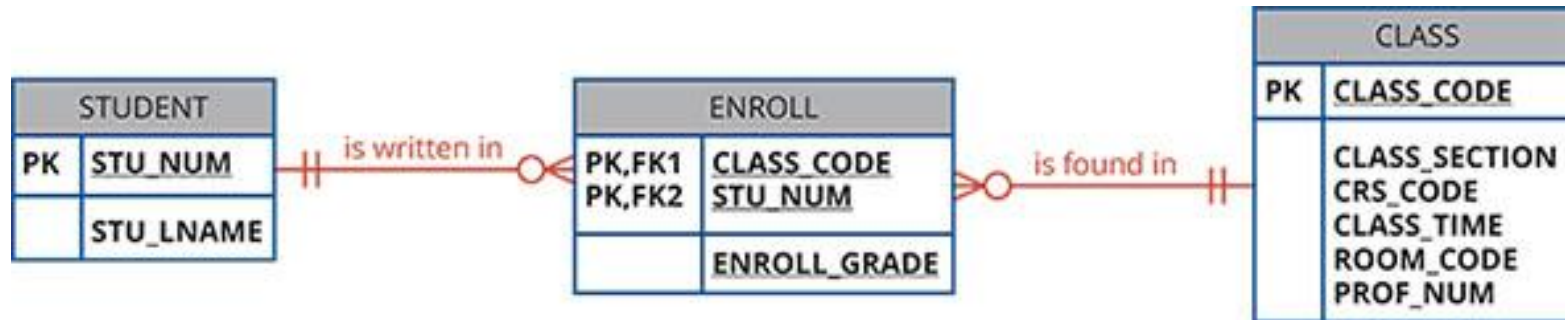
➤ Implementation of the 1:M Recursive Relationship “EMPLOYEE manages EMPLOYEE”

Associative Entities (1 of 2)

- An **associative entity** is used to resolve an **M:N (many-to-many) relationship**.
- It is also called a **composite entity** or **bridge entity**.
- It connects the two original entities using **two 1:M relationships**.
- It usually contains the **primary keys of both parent entities**.
- It can also contain **additional attributes about the relationship**.

Example: Associate Entity

STUDENT and CLASS have an M:N relationship. ENROLL resolves it.



M:N relationship → two 1:M relationships

- A student can enroll in many classes.
- A class can have many students.
- **ENROLL** connects them and stores information such as **ENROLL_GRADE**.

From Business Rules to an ERD

PROCESS

1

Read the scenario

Highlight nouns, verbs, quantities, and constraints.

2

Identify entities

Choose the important nouns that need stored data.

3

Identify relationships

Turn business verbs into relationship names.

4

Set cardinality & participation

Decide minimum and maximum on both sides.

5

Add attributes & identifiers

Choose PKs, required/optional attributes, and domains.

6

Review against every rule

Check that the ERD can express the original scenario.

Tiny College — Business Rules (1 of 2)

BUSINESS RULES

Before drawing the ERD, translate the scenario into clear rules.

1

A SCHOOL operates many DEPARTMENTS; each DEPARTMENT belongs to one SCHOOL.

2

A DEPARTMENT offers many COURSES; each COURSE belongs to one DEPARTMENT.

3

A COURSE generates many CLASSES; each CLASS belongs to one COURSE.

4

A SEMESTER includes many CLASSES; each CLASS occurs in one SEMESTER.

5

A DEPARTMENT employs many PROFESSORS; each PROFESSOR works for one DEPARTMENT.

6

A PROFESSOR may be dean of a SCHOOL; each SCHOOL has one dean.

7

A PROFESSOR may chair a DEPARTMENT; a DEPARTMENT may have one chair.

Tiny College — Business Rules (2 of 2)

BUSINESS RULES

Read every relationship in both directions.

8

A PROFESSOR teaches many CLASSES; each CLASS is taught by one PROFESSOR.

1

A STUDENT can enroll in many CLASSES, and a CLASS can have many STUDENTS.

9

A PROFESSOR advises many STUDENTS; a STUDENT may have one advisor.

1

A BUILDING contains many ROOMS; each ROOM belongs to one BUILDING.

1

A DEPARTMENT has many STUDENTS; a STUDENT may belong to one DEPARTMENT.

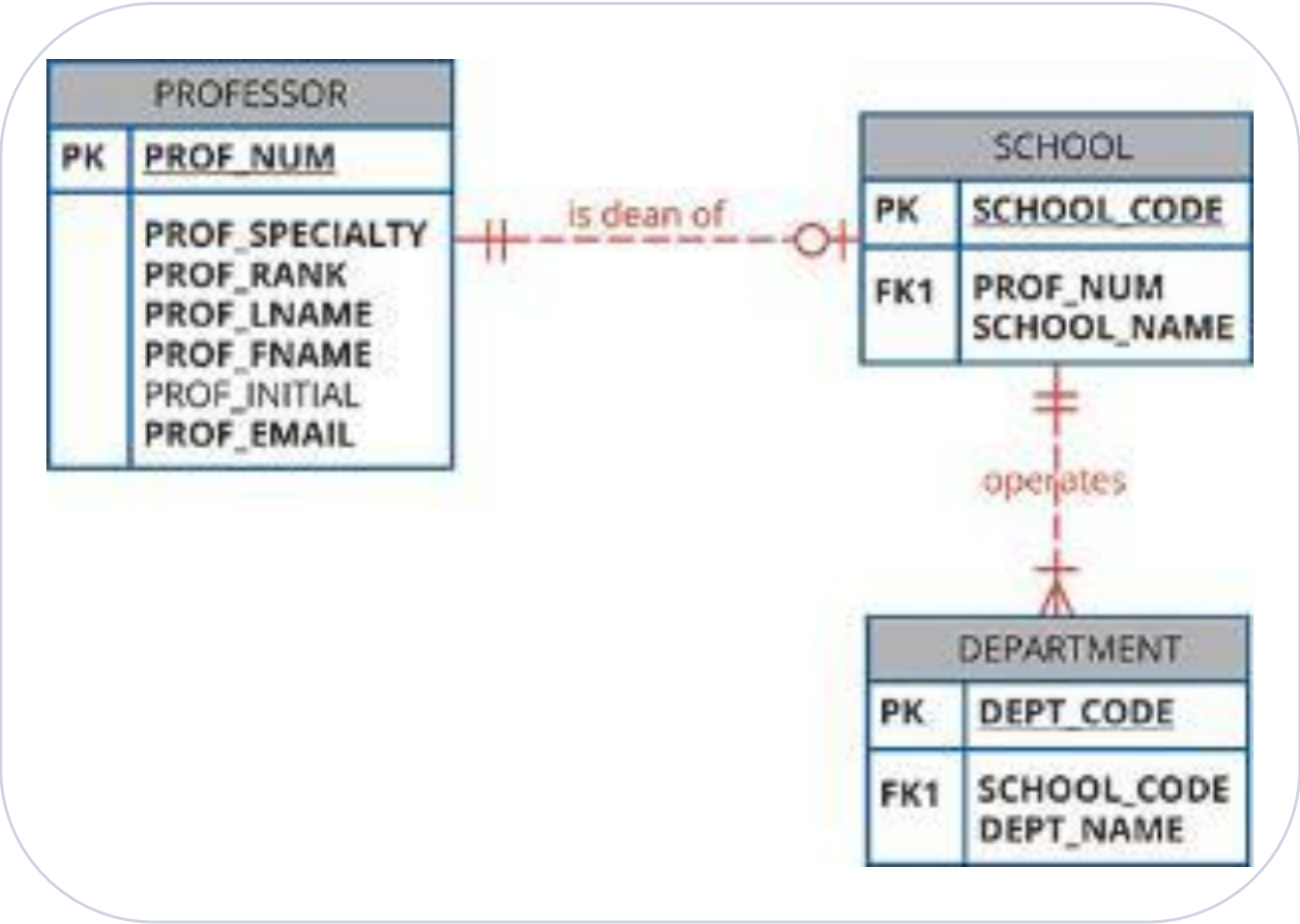
1

A ROOM can be used for many CLASSES; each CLASS uses one ROOM.

Important: STUDENT ↔ CLASS is M:N, so it will need an associative entity.

Step 1: Model SCHOOL, DEPARTMENT, and PROFESSOR

STEP 1



Business rule(s)

- A SCHOOL operates many DEPARTMENTS; each DEPARTMENT belongs to one SCHOOL.
- A PROFESSOR may be dean of a SCHOOL; each SCHOOL has one dean.

Start with the main organizational entities and place the foreign keys on the related side.

Step 2: Add COURSE

STEP 2



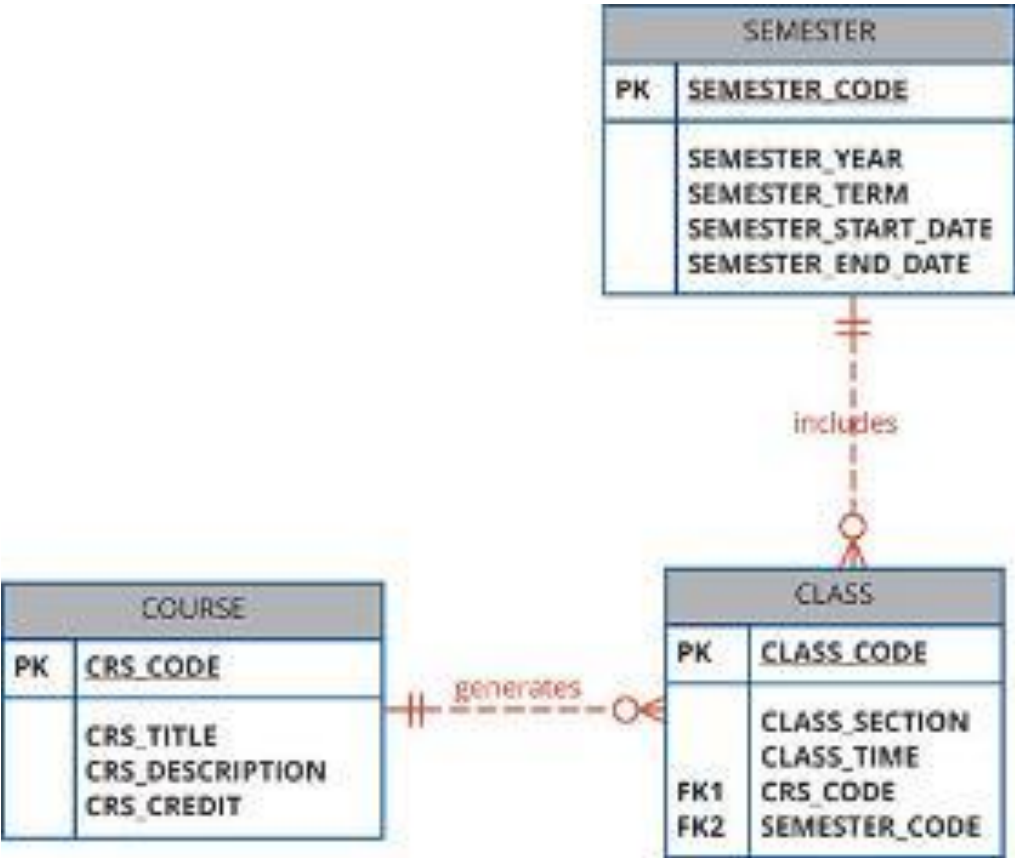
Business rule(s)

- A DEPARTMENT offers many COURSES.
- Each COURSE belongs to one DEPARTMENT.

This is a 1:M relationship. DEPT_CODE appears as a foreign key in COURSE.

Step 3: Add SEMESTER and CLASS

Build the ERD one business rule at a time.



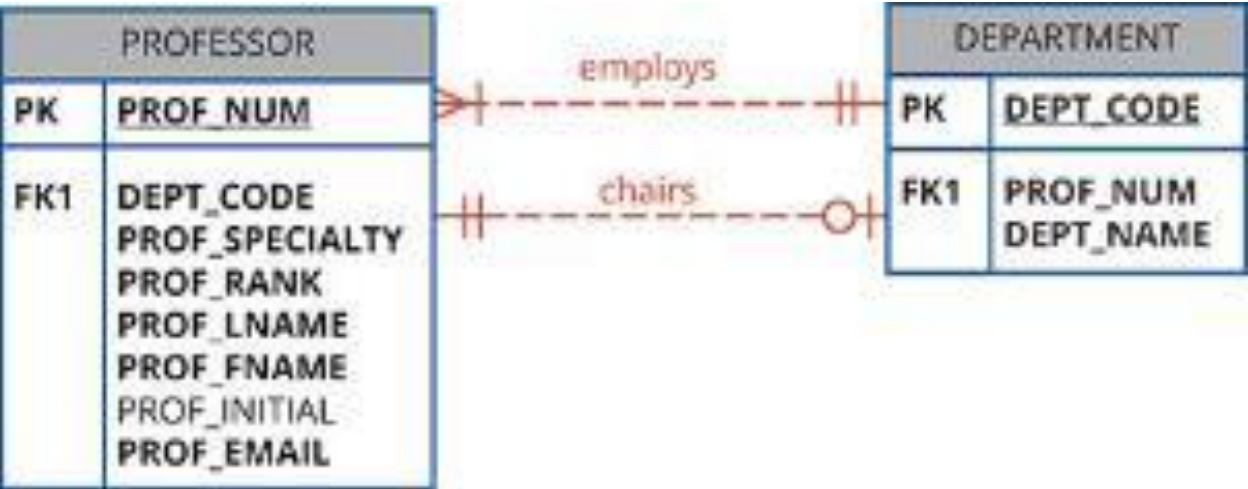
Business rule(s)

- A COURSE generates many CLASSES; each CLASS belongs to one COURSE.
- A SEMESTER includes many CLASSES; each CLASS occurs in one SEMESTER.

CLASS connects to both COURSE and SEMESTER through foreign keys.

Step 4: Add PROFESSOR-DEPARTMENT Relationships

Build the ERD one business rule at a time.



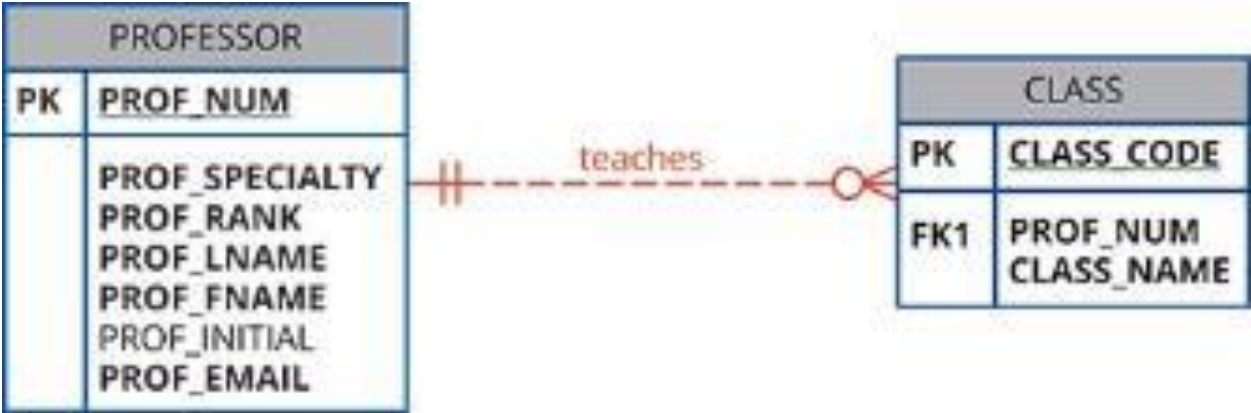
Business rule(s)

- A DEPARTMENT employs many PROFESSORS; each PROFESSOR works for one DEPARTMENT.
- A PROFESSOR may chair a DEPARTMENT; a DEPARTMENT may have one chair.

The same two entities can participate in more than one relationship when the meanings are different.

Step 5: Add Teaching Assignments

Build the ERD one business rule at a time.



Business rule(s)

- A PROFESSOR teaches many CLASSES.
- Each CLASS is taught by one PROFESSOR.

PROF_NUM is stored in CLASS as the foreign key for the teaching relationship.

Step 6: Resolve STUDENT-CLASS M:N with ENROLL

STEP 6

Build the ERD one business rule at a time.



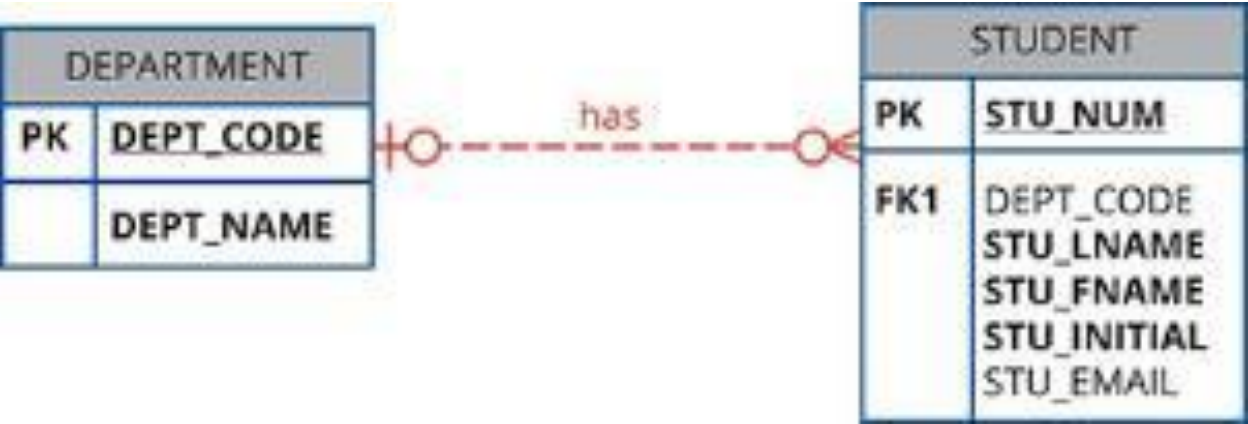
Business rule(s)

- A STUDENT can enroll in many CLASSES.
- A CLASS can have many STUDENTS.

ENROLL converts the M:N relationship into two 1:M relationships and can store EnrollDate and EnrollGrade.

Step 7: Add STUDENT Department

Build the ERD one business rule at a time.



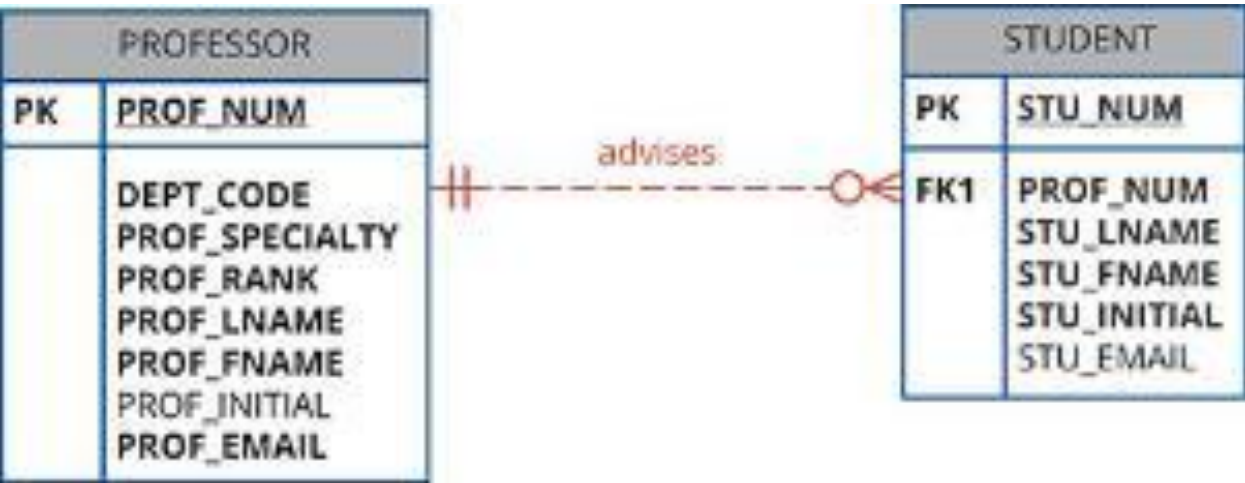
Business rule(s)

- A DEPARTMENT has many STUDENTS.
- A STUDENT may belong to one DEPARTMENT.

DEPT_CODE is stored in STUDENT as a foreign key.

Step 8: Add Student Advising

Build the ERD one business rule at a time.



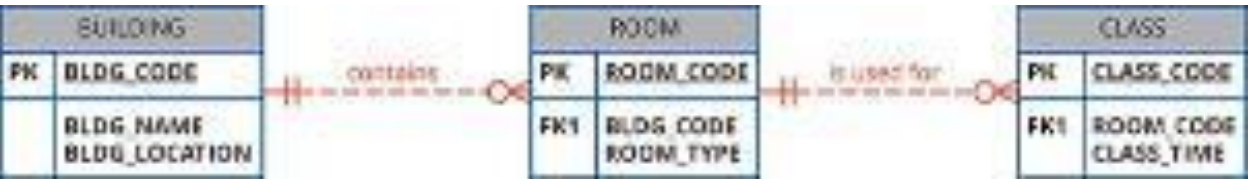
Business rule(s)

- A PROFESSOR can advise many STUDENTS.
- A STUDENT may have one PROFESSOR as an advisor.

PROF_NUM in STUDENT represents the advisor relationship.

Step 9: Add BUILDING and ROOM

Build the ERD one business rule at a time.



Business rule(s)

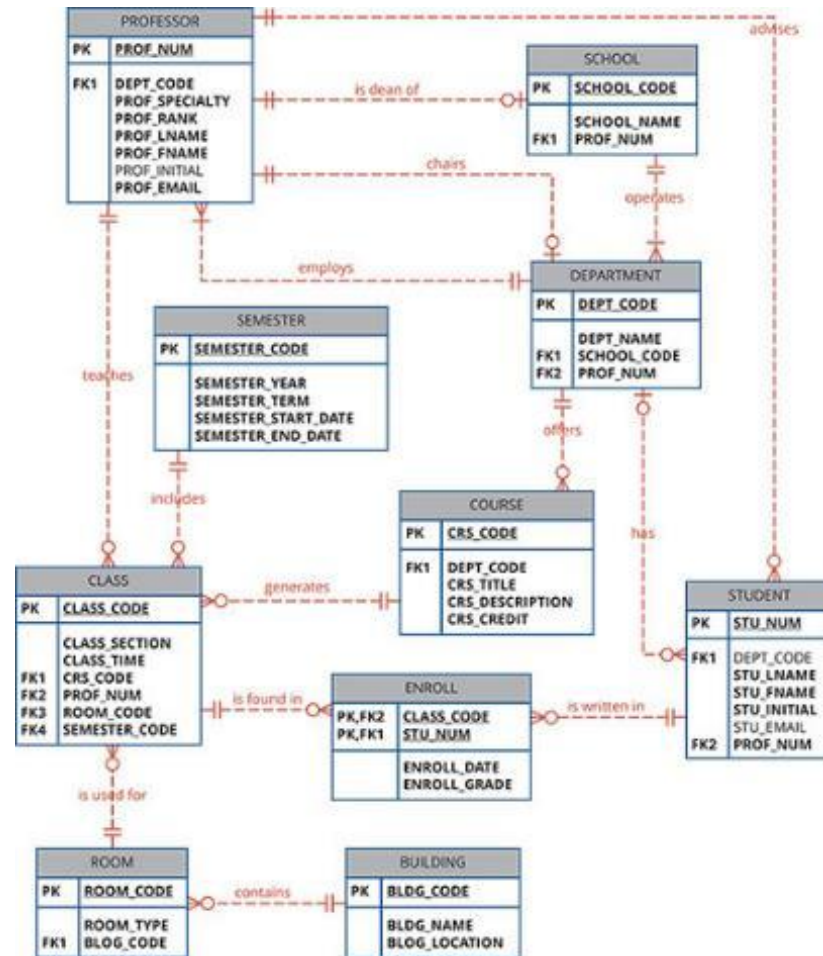
- A BUILDING contains many ROOMS; each ROOM belongs to one BUILDING.
- A ROOM can be used for many CLASSES; each CLASS uses one ROOM.

The model now includes physical resources as well as academic entities.

Final ERD: Tiny College

FINAL ERD

All nine ERD segments combined into one conceptual model.



Final check

- ✓ Every entity has a primary key.
- ✓ Foreign keys implement the relationships.
- ✓ Cardinality matches the business rules.
- ✓ The STUDENT–CLASS M:N relationship is resolved by ENROLL.
- ✓ Relationship names describe the business meaning.

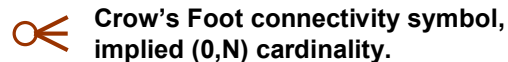
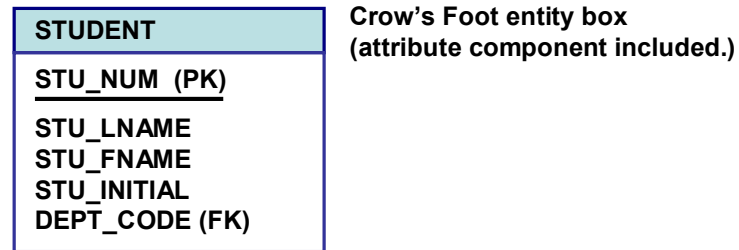
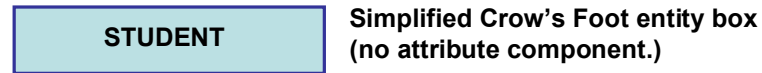
Business rules → ERD → Tables

Knowledge Check Activity

- **How is a relationship between entities indicated in an ERD?**

Knowledge Check Activity 4-2: Answer

- **How is a relationship between entities indicated in an ERD?**
- **Answer:** A relationship is indicated by a line connecting the related entities.



Summary

Now that the lesson has ended, you should be able to:

1. Identify the main characteristics of entity relationship components
2. Describe how relationships between entities are defined
3. Explain how ERD components affect database design and implementation
4. Describe how real-world database design often requires the reconciliation of conflicting goals

Practice

PRACTICE

Apply the ideas before we move to implementation.

Scenario: Library

A MEMBER can borrow many BOOK copies. A physical BOOK_COPY belongs to one BOOK title. Each LOAN records the member, the copy, the checkout date, and the return date.

Your task

- Identify the entities.
- Choose the associative/transaction entity.
- Set minimum and maximum cardinality on each relationship.
- Identify at least one primary key for each entity.